

**Making Signals Naturally Difficult to
Classify with Supervised Learning
Algorithms via Model-Agnostic
Adversarial Perturbations**

UTD Team-1
Midterm Presentation



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Agenda

- Introduction
- Schedule
- Previous Work/Literature Review
- Progress
- Results
- Issues
- Future Work
- Questions

INTRODUCTION



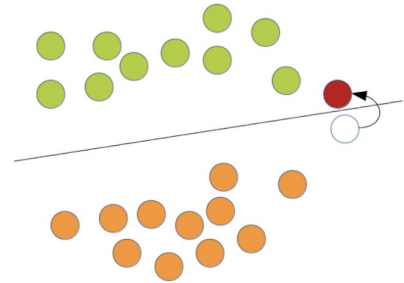
Data Evasion

Most common attack on machine learning model performed during inference.

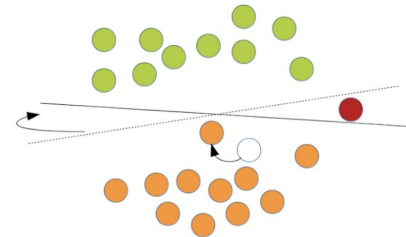


Data Poisoning

The attacker can exploit the ML model by poisoning the data before sending the data to the model for training.



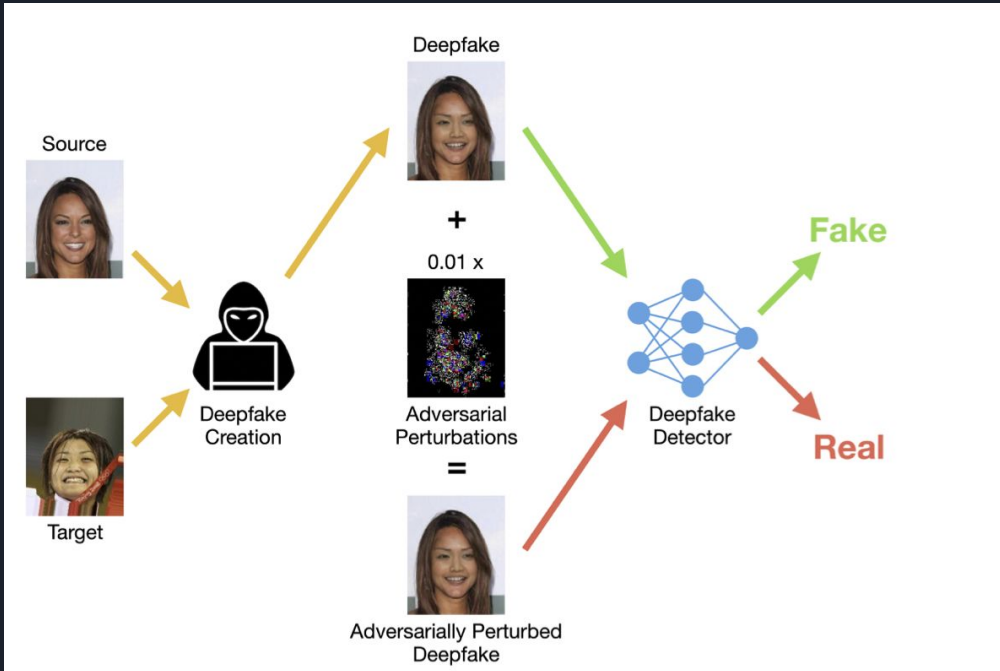
Classical adversarial attack:
directly modifying the testing sample



Data poisoning:
modifying training samples intelligently

INTRO CONT.

Adversarial attacks involve generating slightly perturbed(noise) versions of the input data that fool the classifier(i.e, change its output)



Example of creating a deepfake from source and target faces and adding adversarial perturbations to it. A deepfake detector correctly classifies the deepfake as fake but classifies the adversarially perturbed deepfake as real.



EXPLAINING ADVERSARIAL EXAMPLES

- $\tilde{x} = x + \eta$ where x is the input, $\tilde{x}$ is the adversarial input, η is the perturbation
- Expect same class prediction for x and $\tilde{x}$ if $\|\eta\| < \epsilon$, where ϵ is small enough to be discarded by the sensor or data storage apparatus associated with our problem.
- $w \cdot \tilde{x} = w \cdot x + w \cdot \eta$ w = weight vector, activation to grow by $w \cdot \eta$
- If w has n dimensions and the average magnitude of an element of the weight vector is m , then the activation will grow by mn
- Shows that a simple linear model can have adversarial examples if its input has sufficient dimensionality
- Neural networks linear behavior suggests that cheap, analytical perturbations of a linear model should also damage neural networks.



EXPLAINING ADVERSARIAL EXAMPLES CONT

- Adversarial example generated for one model is often misclassified by other models
- It is seen that multiple classifiers assign the same class to adversarial examples.
- Adversarial examples occur in contiguous regions of the 1-D subspace defined by the fast gradient sign method. Different models learn similar functions when trained to perform the same task.
- The direction of perturbation, rather than the specific point in space, matters most.
- Because it is the direction that matters most, adversarial perturbations generalize across different clean examples.



SCHEDULE

Phase 1: Get a basic understanding of Adversarial Perturbations and fundamentals

Phase 2: Know how to create a substitute model and how to craft adversarial samples

Phase 3: Attack the target model with these adversarial samples



PREVIOUS WORK/LITERATURE REVIEW

References:

1. Flowers, Bryse, R. Michael Buehrer, and William C. Headley. "Evaluating adversarial evasion attacks in the context of wireless communications." IEEE Transactions on Information Forensics and Security 15 (2019): 1102-1113.
2. Practical black-box attacks against machine learning.
<https://dl.acm.org/doi/pdf/10.1145/3052973.3053009?ref=https://githubhelp.com>
3. Explaining and Harnessing Adversarial Examples.
<https://arxiv.org/pdf/1412.6572.pdf>
4. Query-efficient hard-label black-box attack: An optimization-based approach.



PROGRESS

Researched on different attack techniques:

- LBFGS

In target problem, L-BFGS Attack calculated approximate values of adversarial examples by line-searching with $c > 0$

We repeatedly solve this optimization problem for multiple values of c , adaptively updating c

- FGSM

FGSM attack is a *white-box* attack with the goal of *misclassification*

The fast gradient sign method uses the gradient of the loss function to determine in which direction the pixel's intensity should be changed (whether it should be increased or decreased) to minimize the loss function; then, it shifts all pixels simultaneously

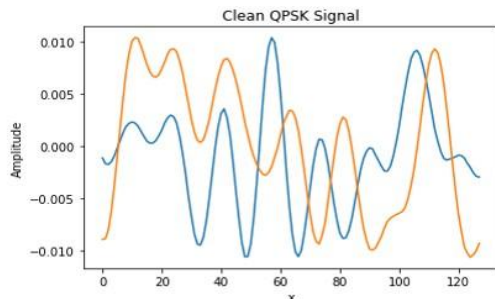
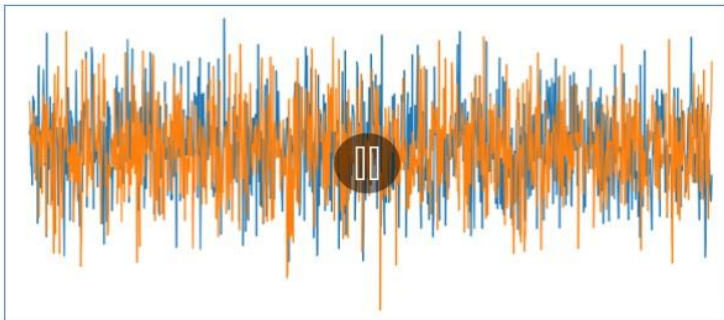
Difference Between them:

- FGSM is designed primarily to be fast instead of producing very close adversarial examples

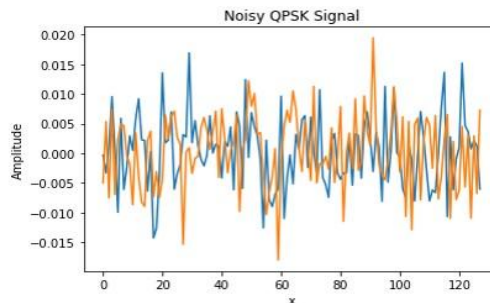
CHALLENGES

DeepSig Dataset ,2018

```
classes = ['32PSK',  
'16APSK',  
'32QAM',  
'FM',  
'GMSK',  
'32APSK',  
'OQPSK',  
'8ASK',  
'BPSK',  
'8PSK',  
'AM-SSB-SC',  
'4ASK',  
'16PSK',  
'64APSK',  
'128QAM',  
'128APSK',  
'AM-DSB-SC',  
'AM-SSB-WC',  
'64QAM',  
'QPSK',  
'256QAM',  
'AM-DSB-WC',  
'OOK',  
'16QAM']
```

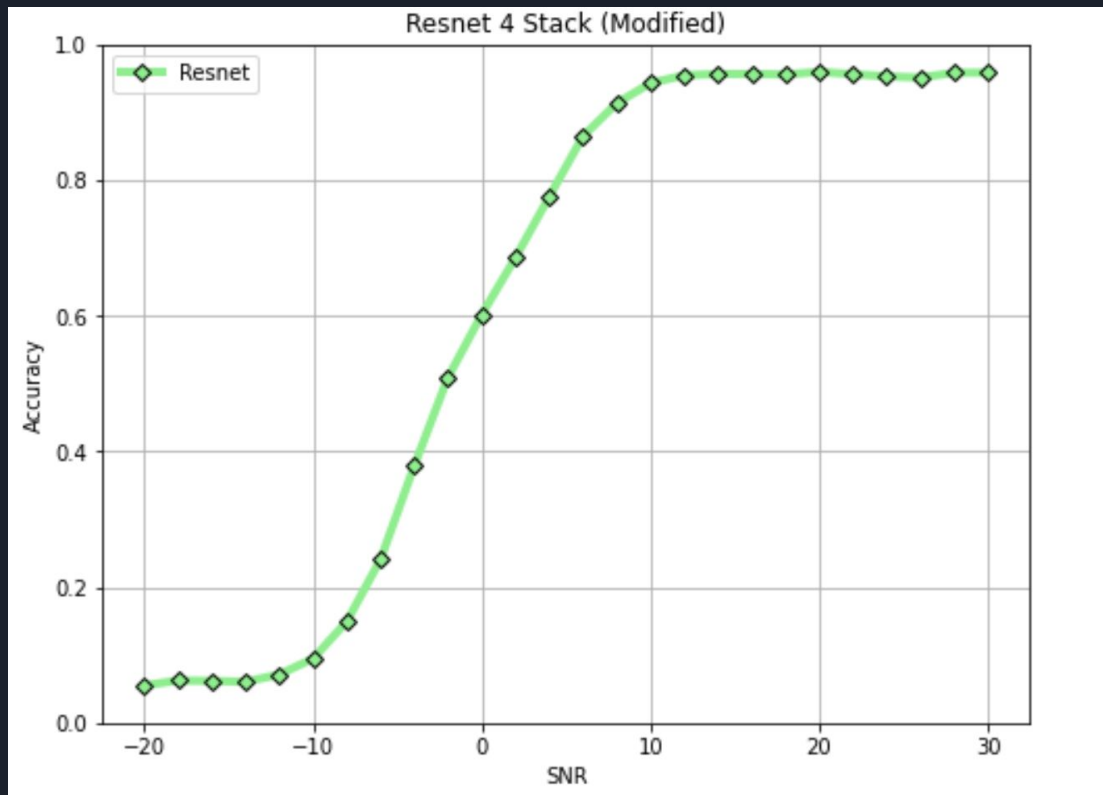


High VS Low
SNR

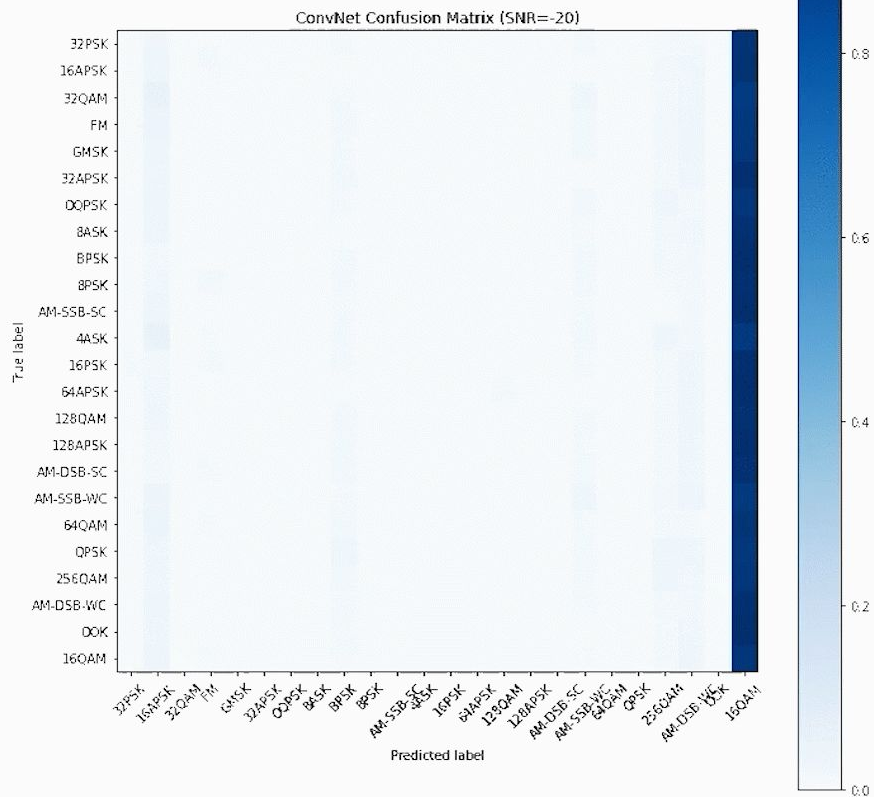


1. In the image below, we can see how drastically noise can affect our ability to recognize a signal.
2. We should be able to classify signals both in low SNR and high SNR.
3. We use the DeepSig Radio Signal Dataset, this dataset is pretty large (18GB). We will use 1.5 Million variation of signal modulation with respect to the SNR value.

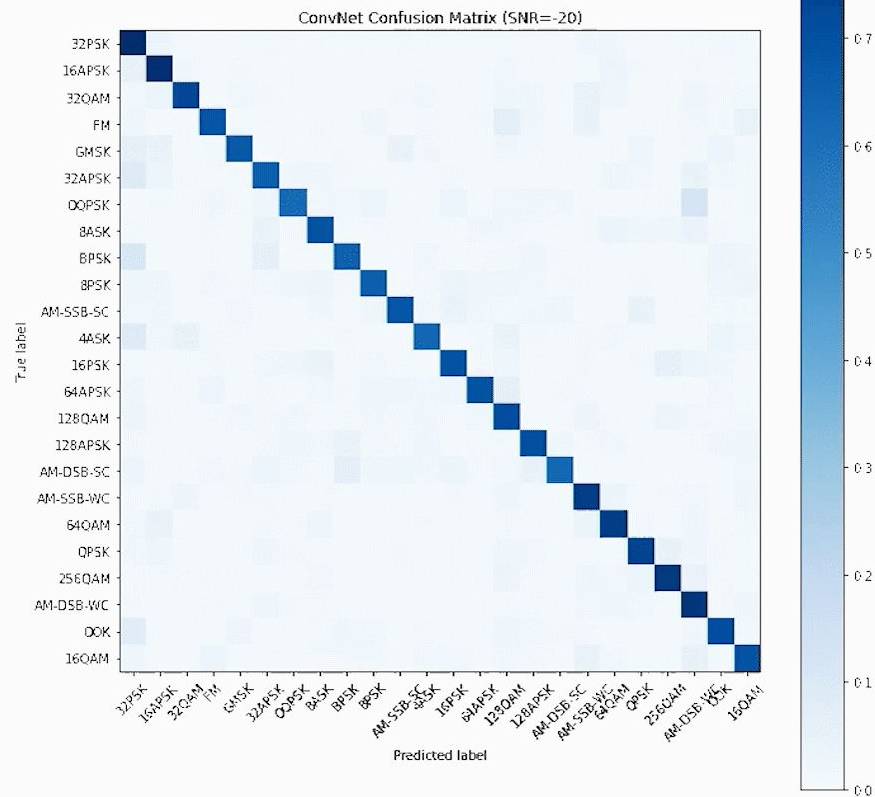
Baseline Model



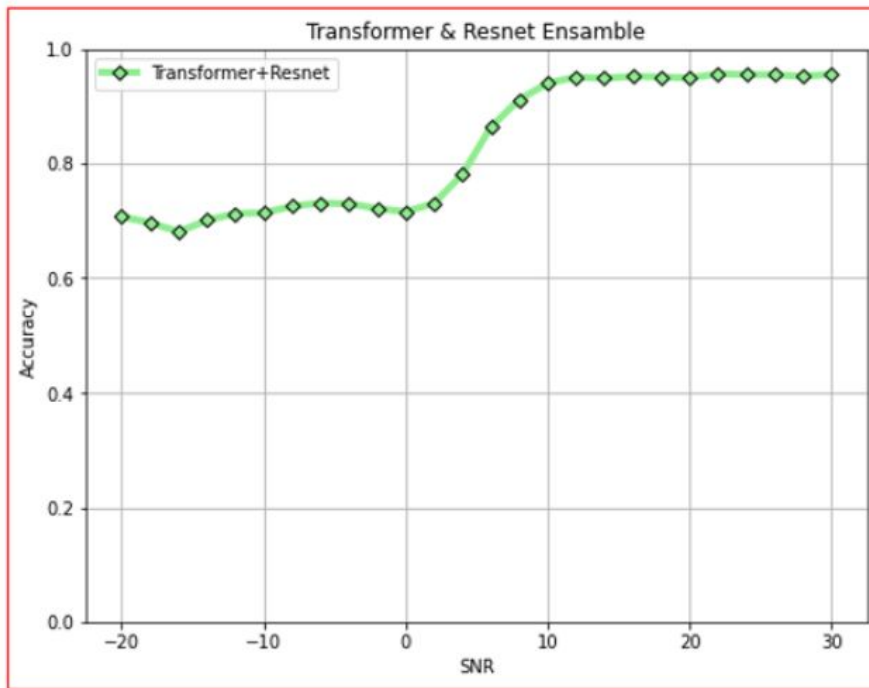
Baseline



Our Method



Our Model





Challenges faced

- Tried on multiple systems to run the data set as it is 21 gb large
- Resolution of errors
 - Huge barrier to train signal data is vanishing gradients



Future Work

1. We want to identify the most powerful attack that can use (most incorrect classifications for a given power constraint) to attack an unknown, un-queryable classifier.
2. After identifying the attack, we now validate our attack on RML 2018 dataset forcing it to misclassify.
3. Once we validate our attack model, then document this study and publish using some publication framework.



QUESTIONS

